Paleomagnetic history of the Mealy dykes of Labrador, Canada[1]

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Paleomagnetic analysis of the Mealy diabase dykes of Labrador reveals magnetizations that predate the Grenville event at about 1000 Ma. These dykes intrude the Mealy Mountains anorthositic complex in the Grenville Structural Province. They are well south of the Grenville Front Tectonic Zone, but were apparently never subjected to temperatures as high as 500°C during their post-consolidation history.

Four distinct magnetic components were uncovered by thermal and alternating field treatments and a fifth remained unresolved. The major magnetic mineral present, titanomagnetite, is thought to record two magnetic directions acquired during cooling from magmatic temperatures. These are *B* ($D = 305°$, $I = -76°$; $N = 18$ sites; $\kappa = 12$; $\alpha_{95} = 11°$) and *A* ($D = 095°$, $I = +52°$; $N = 20$ sites; $\kappa = 46$; $\alpha_{95} = 5°$). Component *B* has much within-site dispersion. The other two components, *C* ($D = 274°$, $I = -47°$; $N = 10$ sites; $\kappa = 15$; $\alpha_{95} = 13°$) and *D* ($D = 292°$, $I = -74°$; $\kappa = 5$; $\alpha_{95} = 31°$), probably reside in magnetite and pyrrhotite, respectively. Component *C*, antiparallel to *A*, was probably acquired at about the same time as *A*. We suggest that *C* and *A* represent the first stable magnetizations retained by the dykes following an extensive period of cooling and re-equilibration of the magnetic minerals. Components *B* and *D*, which agree in direction, represent a later stage of cooling.

Component *B* has a pole at 148°E, 34°S ($\delta p = 18°$, $\delta m = 19°$) in agreement with regional metamorphic poles from the Grenville; *A*, however, has a pole at 173°W, 23°S ($\delta p = 5°$, $\delta m = 7°$), which apparently "sees through" the peak in Grenville activity. The *A* site poles have a linear distribution along the Keweenawan Track and probably relate to an age between 1000 and 1150 Ma.

L'étude paléomagnétique des dykes de diabase de Mealy au Labrador indiquent des aimantations qui antidatent l'évènement du Grenville à environ 1000 Ma. Ces dykes pénètrent dans le complexe anorthositique des montagnes Mealy de la province structurale du Grenville. Ils sont localisés bien au sud de la zone tectonique du front du Grenville, mais selon toute apparence ils n'ont pas été soumis à des températures pouvant dépasser 500°C depuis leur consolidation.

Quatre composantes magnétiques distinctes sont déduites des traitements thermiques et de champ alternatif et une cinquième demeure non-résolue. Le principal mineral magnétique présent, la titanomagnétite, semble porter deux directions magnétiques acquises lors du refroidissement magmatique. Elles sont *B* ($D = 305°$, $I = -76°$; $N = 18$ sites: $\kappa = 12$; $\alpha_{95} = 11°$) et *A* ($D = 095°$, $I = +52°$; $N = 20$ sites; $\kappa = 46$; $\alpha_{95} = 5°$). La composante *B* est bien à l'intérieur des sites de dispersion. Les deux autres composantes *C* ($D = 274°$, $I = -47°$; $N = 10$ sites; $\kappa = 15$; $\alpha_{95} = 13°$) et *D* ($D = 292°$, $I = -74°$; $\kappa = 5$; $\alpha_{95} = 31°$) sont probablement portées respectivement par la magnétite et la pyrrhotite. La composante *C*, antiparallèle à *A*, fut probablement acquise plus ou moins simultanément à *A*. Nous suggérons que *C* et *A* représentent les premières aimantations stables retenues par les dykes suivies d'une longue période de refroidissement et de rééquilibration des minéraux magnétiques. Les directions de *B* et *D* s'accordent et elles représentent une étape ultérieure de refroidissement. Le pôle de *B* se situe à 148°E, 34°S ($\delta p = 18°$, $\delta m = 19°$) conformément aux pôles métamorphiques du Grenville; le pôle *A*, cependant, est à 173°O, 23°S ($\delta p = 5°$, $\delta m = 7°$), et apparemment "voit au travers" le paroxisme de l'activité du Grenville. Les pôles du site *A* présentent une distribution linéaire le long de la courbe du Keweenawien et correspondent possiblement à un âge entre 1000 et 1150 Ma.

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Introduction

The purpose of this study was to determine paleopoles (geomagnetic field directions) that predate the widespread 1000 Ma thermal event characteristic of Grenville terrane.

Paleomagnetic studies within the Grenville Structural Province have until now revealed two general groups of poles joined in a smeared distribution. One group, the so-called "A" poles (G(A), see Fig. 13), is thought to be associated with magnetizations that originated at the time of the Grenvillian regional metamorphism; the other group is the so-called "B" poles (G(B)), with magnetizations related to later uplift and cooling. These data have provided a basis for several models of Grenville evolution: one envisioning the partial reactivation of a single plate (Roy and Fahrig 1973) and the other envisioning a collision between two plates (Irving *et al.* 1972). Current evidence favours the first model with the G(A) and G(B) poles simply representing different stages of the cooling history. But these models

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only concern post-Grenvillian Orogeny history; the earlier history has been hidden from paleomagnetic probing by the pervasive metamorphism. The recovery of pre-Grenville magnetizations would constrain future plate tectonic models of the Grenville Province.

Paleomagnetic studies have generally been confined to the easily accessible southwestern portion of the Grenville Province, where a high grade of metamorphism prevails. Two areas, the Hastings Lowlands of southeastern Ontario, and the northeastern part of the Grenville Province in Labrador, have significantly lower grades of metamorphism, and thus provide better opportunities for recovering primary magnetizations (Dunlop *et al.* 1980). In the Hastings Lowlands, paleomagnetic studies of the Thanet gabbros have offered equivocal evidence of the presence of primary magnetization (Buchan 1978; Dunlop *et al.* 1980). Previous studies in Labrador on units along the Grenville Front (Fahrig and Larochelle 1972; Fahrig *et al.* 1974) have also yielded somewhat equivocal results, possibly due to inadequate analysis. The present paleomagnetic study from the Mealy dykes of Labrador, located well south of the Grenville Front, provided a good opportunity for recovering a primary magnetization.

Geology and geochronology

The Mealy dykes constitute a concentrated olivine diabase swarm transecting anorthositic and monzonitic rocks of the Mealy Mountains complex (Emslie 1976). The region lies well within the Grenville Structural Province (Fig. 1) but, perhaps surprisingly, little or no deformation or regional metamorphism directly attributable to the Grenvillian Orogeny is displayed in either the host rocks forming the Mealy Mountains complex or the dykes themselves.

The Mealy dykes lie within a zone more than 200 km wide of prominent Neohelikian hypabyssal and volcanic basic magmatism that straddles the Grenville Front Tectonic Zone and trends east-northeast for more than 600 km (Fig. 1). As pointed out by Meyers and Emslie (1977), other basaltic and gabbroic suites within the zone that appear to be roughly contemporaneous include the Shabogamo gabbros, the Harp dykes, the Michael gabbros, and the Seal Lake Group basalts and gabbros (Fig. 1, Table 1).

The Mealy dykes were intruded some 250 Ma after the cooling and perhaps considerable uplift of the Mealy Mountains complex. They have been dated by the Rb–Sr technique at 1380 ± 54 Ma (Table 1). Whole-rock K–Ar dyke ages of 1078 ± 48, 1123 ± 50, and 1222 ± 101 Ma (Table 1) may be interpreted, individually and collectively, as time estimates of whole-rock closure to radiogenic argon diffusion in dykes, although they cannot be precisely related to temperatures. Biotite ages of 955 and 964 Ma can be interpreted as uplift–cooling ages probably related to Grenvillian tectonism. They presumably record cooling of the dykes through an Ar blocking temperature of about 300°C.

Sites were collected by one of us (RFE) along the shores of Lake Melville (14 sites) and in the generally well exposed higher terrane of the Mealy Mountains (8 sites) (Fig. 2). Sampled dykes range from about 7 to 75 m in width.

Methods

Core samples were drilled in the field, generally from the interiors of the dykes, and oriented by sun and magnetic compasses (orientation errors are estimated to be $<5°$). Cores were cut into two standard specimens (2.2 cm in height) in the laboratory. Experiments were then performed on specimens using either alternating field (AF) treatment, thermal demagnetization, or a combination of the two methods in order to analyze the natural remanent magnetization (NRM). Measurements of the remanent magnetization after each predetermined field or temperature step were variously made using Schonstedt spinner magnetometers (DSM-1 or SSM-1A) or a Geofyzika JR 4 spinner. AF experiments were performed using a commercial instrument (Schonstedt with two-axis tumbling) or the three-axis tumbling apparatus described by Roy *et al.* (1973). Thermal demagnetization was conducted within Schonstedt TSD-1 furnaces or the large furnace of Roy *et al.* (1972). Residual fields at the specimen within the non-commercial AF and thermal instruments were usually <2 nT (Roy *et al.* 1969).

The experimental data were analyzed by the aid of vector diagrams (Zijderveld 1967). Clearly defined magnetic components (i.e., straight-line segments) were fitted visually.

Results and analysis

Resolution of directions

NRM's had intensities mainly ranging from 1 to 10 A/m and directions that, though scattered, tended to cluster in the east with positive, intermediate inclinations (Fig. 3).

Analysis of the vector diagrams of experimental data resolved as many as four magnetic components from single specimens. A fifth component remained unresolved. The magnetization generally uncovered first, called *B*, was revealed by AF treatment in the field range up to 9 or 10 mT (Fig. 4*a*) or by thermal treatment in the temperature range up to 400–520°C (Figs. 4*b*, 5–7), and has a steep negative direction (Fig. 8*a*). In some instances below about 5 mT another magnetization, *D*, was clearly revealed first (e.g., Fig. 6), but in most cases *B* and *D* had apparently overlapping resistive coercive force (rcf) ranges. The third magnetization *A* was

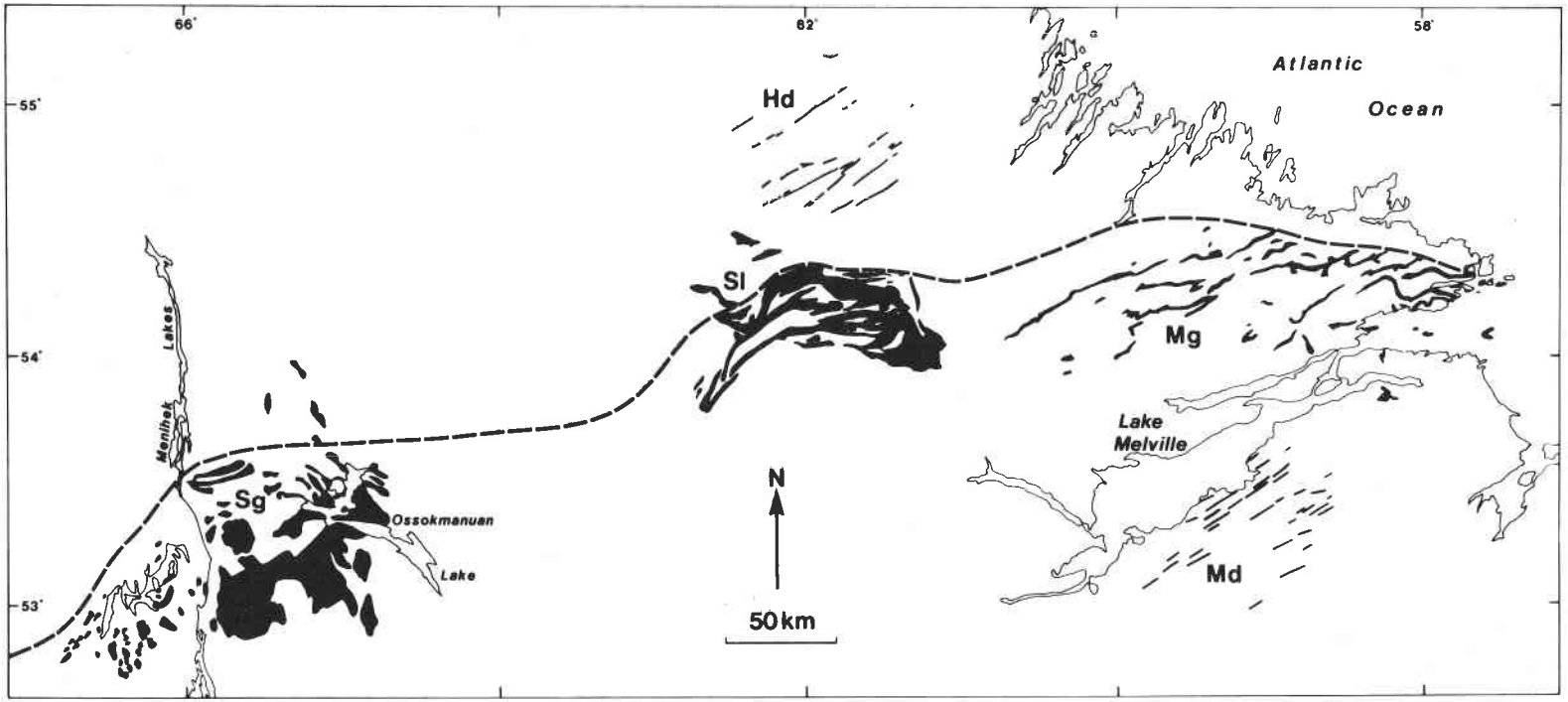


FIG. 1. Regional map of Labrador showing the location of the Mealy dykes (Md) and the possibly related rock units: the Harp dykes (Hd), the Michael gabbro (Mg), the Shabogamo gabbro (Sg), and the Seal Lake Group basalts and diabases (Sl). The dashed line is the approximate northern limit of the Grenville Front Tectonic Zone (e.g., Gower *et al.* 1980).

TABLE 1. Radiometric dates

Age (Ma)	Material dated	Comments	Reference
K–Ar			
964±21	Biotite	Mealy dyke, site EC7535	GSC
955±27	Biotite	Mealy dyke, site EC7505	GSC
1078±48	Whole Rock	Mealy dyke	Gittins (1972)
1222±50	Whole Rock	Mealy dyke	Gittins (1972)
951±88	Hornblende	Amphibolite gneiss in fault contact with monzonite EC75-243A	GSC
1400±57	Hornblende	Amphibolite gneiss inclusion in Etagaulet anorthosite	GSC
1533±45	Hornblende	Igneous hornblende from troctolitic margin of Kenemich anorthosite	GSC
Rb–Sr			
1678±77	Whole rock (7 samples)	Isochron age, monzonite–granite suite, Mealy Mountains complex	Emslie *et al.* (1983)
1380±54	Whole rock (7 samples)	Isochron age, Mealy dykes	Emslie *et al.* (1983)
U–Pb zircon			
1640±3	Zircon	Monzonite–granite suite, Mealy Mountains complex	Emslie *et al.* (1983)
Related			
1323±92	(Rb–Sr, whole rock)	Seal Lake Group basalts and gabbros	Baragar (1981)
1316±94	(K–Ar, whole rock)	Harp dykes	Wanless *et al.* (1978)
1379±69	(Sm–Nd, Rb–Sr, whole rock)	Shabogamo intrusive suite	Zindler *et al.* (1981); Brooks *et al.* (1981)
1461±96	(Rb–Sr, whole rock)	Michael gabbros	Fahrig and Loveridge (1981)

NOTE: GSC references refer to unpublished age determinations by Geological Survey of Canada.

revealed by AF's above 10 mT (Fig. 4*a*) and by thermal demagnetization—sometimes following initial AF treatment to 9 mT or so (Fig. 6)—in the interval between about 400 and 560°C (Figs. 4*b*, 5, 6, 7*b*). It generally has a positive easterly direction of intermediate inclination (Fig. 9*a*). The last resolved magnetic component, *C*, was uncovered mainly by thermal treatment in the range from about 560 to 580°C, and often appears to be antiparallel to *A* (Figs 5 *c* and *d*; 6 *c* and *d*; 7*b*), that is, with a negative westerly direction of intermediate inclination (Fig. 9). Above the range 600–640°C an induced component apparently masks part of the unblocking temperature (T_{UB}) range of *C* (e.g., see Figs. 4*c* and 7*c*). In this range there is positive evidence for the existence of a fifth component, but it could not be resolved (Fig. 7*c*).

Statistical treatment

The *B*, *A*, and *C* directions are plotted in Figs. 8 and 9 and set out in Tables 2, 3, and 4. Component *A* has a distinctly non-spherical distribution of directions, with an eccentricity of 0.93 as calculated by the method of Engebretson and Beck (1978). The *B* distribution is also non-spherical, with an eccentricity of 0.77. Such non-spherical distributions, especially like that of *A*, are more properly characterized by the Bingham (Onstott 1980) rather than the Fisher (Fisher 1953) distribution and statistics. Because a computer program has been unavailable for performing the Bingham calculations, grand averages have been obtained using the Fisher method, but the poles should be correct to within several degrees using this latter method. Possible reasons for the skewed directions are discussed in the next section.

The softest magnetization, *D*, is only well defined in seven specimens (sp). It has an average from rather scattered directions of $D = 292°$, $I = -74°$ ($N = 7$ sp; $\kappa = 5$; $\alpha_{95} = 31°$). The *B* magnetization has a similar average, also from scattered data, from both AF ($N = 74$ sp; $D = 303°$, $I = -77°$; $\kappa = 4$; $\alpha_{95} = 9°$) and thermal ($N = 53$ sp; $D = 315°$, $I = -77°$; $\kappa = 6$; $\alpha_{95} = 9°$) directions. Component *A* was the most common, and generally predominant, magnetization resolved, being found in all sites except 14, and 16, which yielded a crudely reverse direction (Fig. 9*b*). Because the site 16 direction differs somewhat from that of the other sites, it is not included in the *A* grand average. Both the AF and thermally treated specimens yield virtually the same *A* direction (AF: $N = 45$ sp; $D = 096°$, $I = +51°$; $\kappa = 31$; $\alpha_{95} = 4°$ versus thermal: $N = 69$ sp; $D = 096°$, $I = +52°$; $\kappa = 47$; $\alpha_{95} = 3°$), and these results agree quite well with the thermal results of other specimens initially treated by AF ($N = 54$ sp; $D = 100°$, $I = +50°$; $\kappa = 39$;

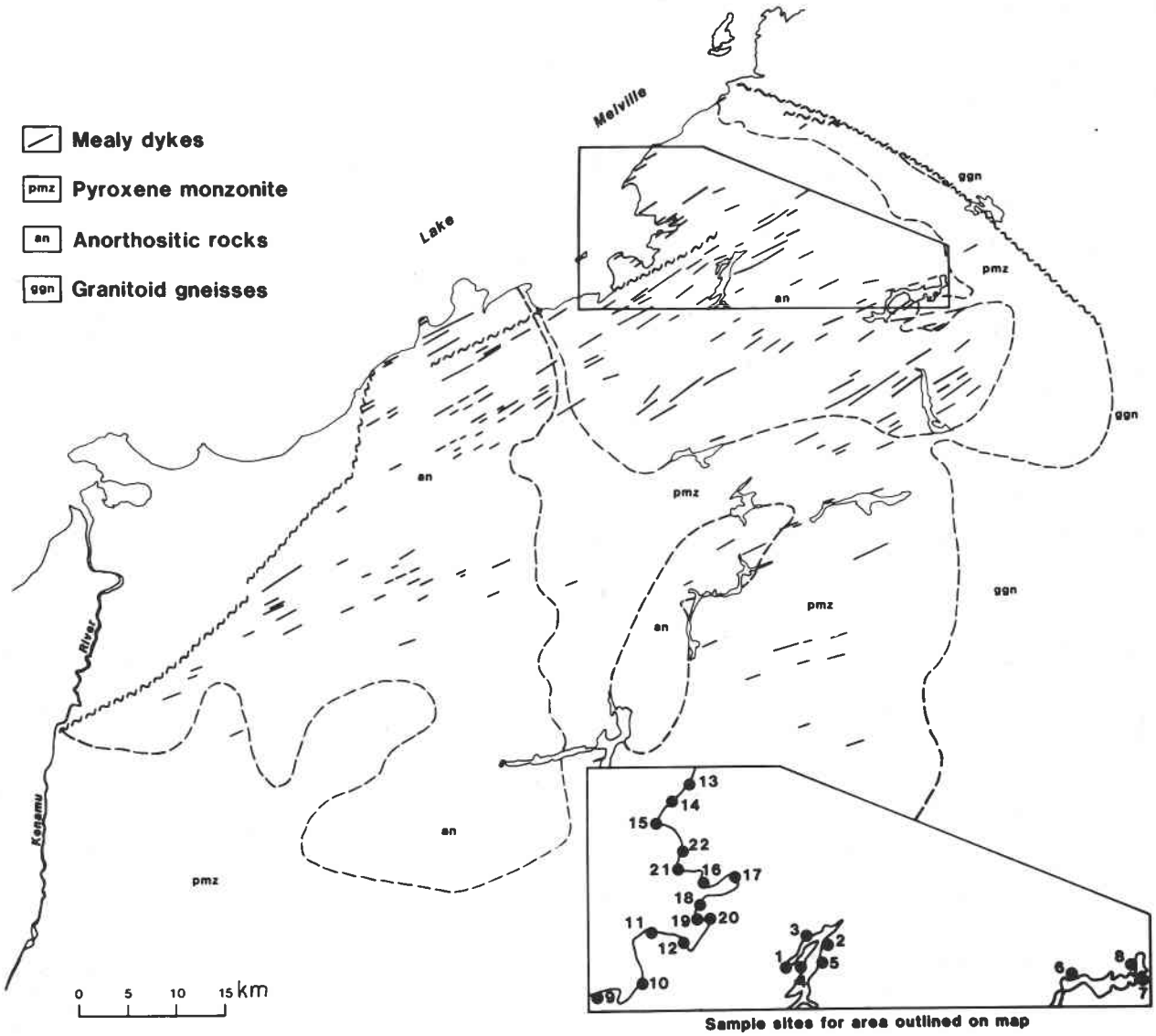


FIG. 2. Geological map of Mealy Mountains area showing the site localities.

$\alpha_{95} = 3°$). AF and thermal data are nearly identical for the *A* determination. The same applies for the *B* determination. Therefore in both cases AF and thermal data were averaged at the specimen level to obtain sample data from which the site averages were calculated (Fig. 8*b*, Table 2; Fig. 9*b*, Table 3). The *C* magnetization was resolved from only 15 samples from 10 sites (Fig. 9*b*, Table 4) and is essentially antiparallel to the *A* direction. The grand averages for all the directions except *D* are given in Table 5.

Relation of directions to magnetic minerals

How do these directions relate to the magnetic minerals present? From a study of the opaque mineralogy by microscope and electron microprobe analysis three magnetic minerals were identified. The predominant magnetic mineral present is titanomagnetite, which generally occurs as lamellar or trellis intergrowths with ilmenite, commonly in grains 0.25–1 mm in size. Together with separate, mostly homogeneous, ilmenite (Ilm) grains of similar size they constitute the dominant oxides by volume. In addition, abundant, regularly exsolved, 1–2 μm plates and rods in clinopyroxenes are mainly or entirely titanomagnetite. The amount of TiO_2 in solid solution in magnetite ranges from 1.5 to 3.9 wt.% over four samples from different sites, or the

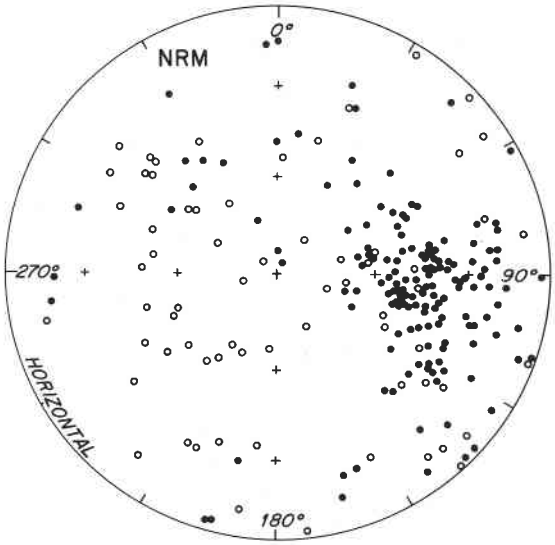


FIG. 3. Natural remanent magnetization directions of all specimens. Solid (open) circles represent positive (negative) inclinations.

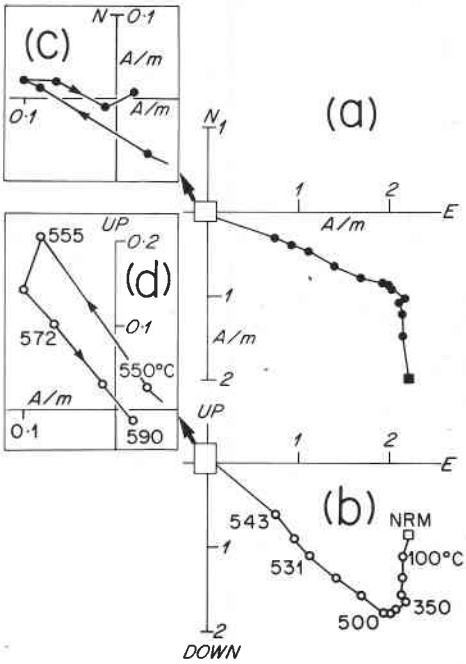


FIG. 5. Vector diagrams depicting the thermal decay of a specimen from site 15 (153A) in the horizontal (*a*) and vertical (*b*) planes. (*c*), (*d*) Expanded views of the origin. See Fig. 4 for other explanations.

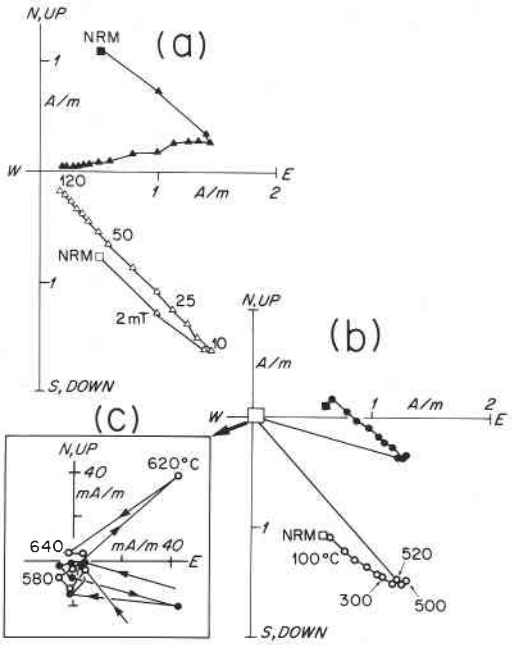


FIG. 4. Vector diagrams of treated companion specimens of a sample from site 8, one (82B) treated by AF (*a*) and the other (82A), thermally (*b*, *c*). In this and succeeding vector diagrams triangles represent AF results, circles thermal results, and squares the NRM's. The solid symbols represent vectors plotted on the horizontal north–east plane and the open symbols vectors plotted on the vertical up–east plane.

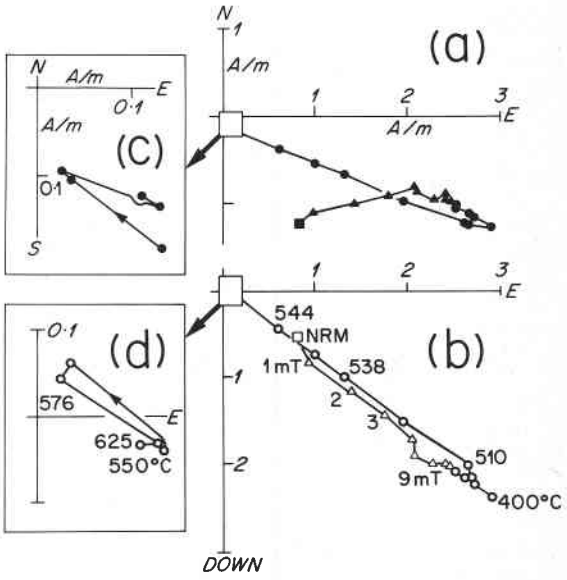


FIG. 6. Vector diagrams in the horizontal (*a*) and vertical (*b*) planes showing the two-stage treatment (AF followed by thermal) of a specimen (151A) from site 15. (*c*), (*d*) Expanded views of the origin. See Fig. 4 for other explanations.

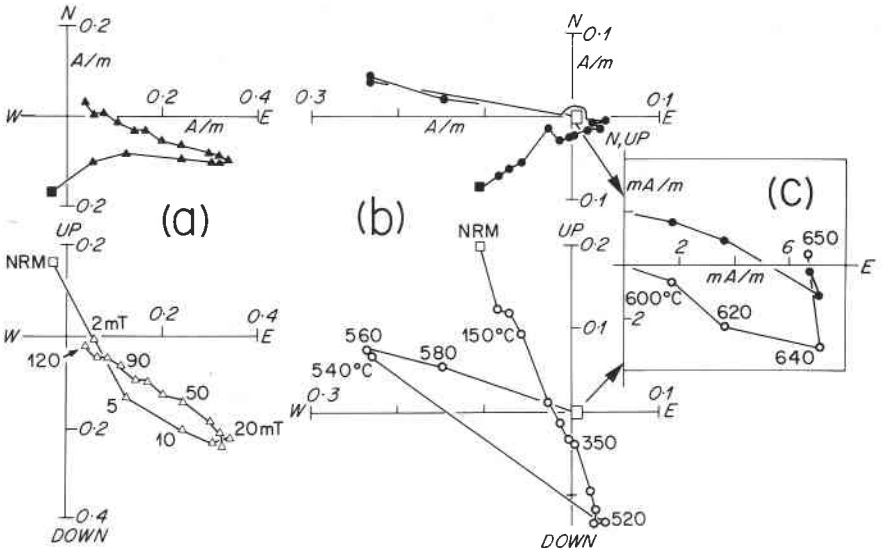


FIG. 7. Vector diagrams of the decay of magnetization of two companion specimens from a sample of site 22, one (222B) treated by AF (*a*) and the other (222A) thermally (*b*). The origin of (*b*) has been expanded in (*c*). See Fig. 4 for other explanations.

equivalent of 4.2–10.9 mol% ulvospinel (Buddington and Lindsley 1964). These percentages translate to Curie points ranging from about 554 to 516°C (McElhinny 1973, p. 42), which are within the T_{UB} range of the *A* component (about 560–520°C). We therefore suggest that titanomagnetite is the carrier of *A*. In many cases *B* has exactly "complementary" rcf and T_{UB} spectra to those of *A*; that is, there is a sharp boundary between the spectra. This fact is consistent with the idea that both magnetizations reside in the same magnetic mineral. In other cases there is apparent overlap of the *A* and *B* spectra. However, on the basis of indirect evidence given below, we suggest that *B* also resides in titanomagnetite.

Another magnetic mineral, (titano)hematite, is readily identified in only a few samples (from sites 12, 16, and 18) that show some degree of deuteric or secondary alteration. In these samples, it typically fills narrow fractures and occurs within cleavages of Fe–Mg silicates. It was also identified in the form of very rare but fine lamellae in ilmenite. Some hematite may additionally occur as 1–2 μm plates and rods in clinopyroxenes and may also account for the cloudy brownish colour of plagioclase. The reduced character of the rock probably accounts for the lack of hematite in the primary mineral assemblages. It is possible that the unresolved fifth magnetization with T_{UB}'s above 580°C resides in the hematite.

The third magnetic mineral identified, pyrrhotite, which is locally quite common, has a primary igneous origin. Soffel (1977) determined that the rcf's of pyrrhotite range from below 10 mT for large multidomain (md) grains, to between 10 and 30 mT for smaller md grains and larger ones having dispersed inclusions, to greater than 100 mT for single domain (sd) grains, those below 1.6 μm. In the present samples, pyrrhotite ranges in size up to 1 mm, thus correlating (in part) with large md grains with rcf's below 10 mT. But both the *B* and *D* components have rcf's below 10 mT. In addition, pyrrhotite has a Curie point of 320°C, which possibly correlates with a minor end point occurring between 300 and 350°C in the *B* unblocking temperature range in a number of vector diagrams (e.g., Figs. 4*b*; 5 *a* and *b*; 7*b*). Thus pyrrhotite may harbour either the *D* component or the *B* component in part. This matter is further discussed in the next section.

The *C* magnetization with T_{UB}'s up to 580°C probably resides in nearly pure magnetite. Magnetite (Mt) formed by several processes during dyke history. Ulvospinel-rich Mt_{ss} and hematite-poor Ilm_{ss} formed part of the primary crystallized mineral assemblage of the dykes; oxidation–exsolution processes produced increasingly purer magnetites with rising Curie temperatures as cooling proceeded. Of quite different origin is magnetite of unknown titanium (Ti) content that occurs as fine-grained exsolution products in pyroxenes and plagioclase.

Discussion and conclusions

Sequence of magnetization

Paleomagnetic analysis has resolved four distinct magnetic components: *A*, *B*, *C*, and *D*. Components *A*

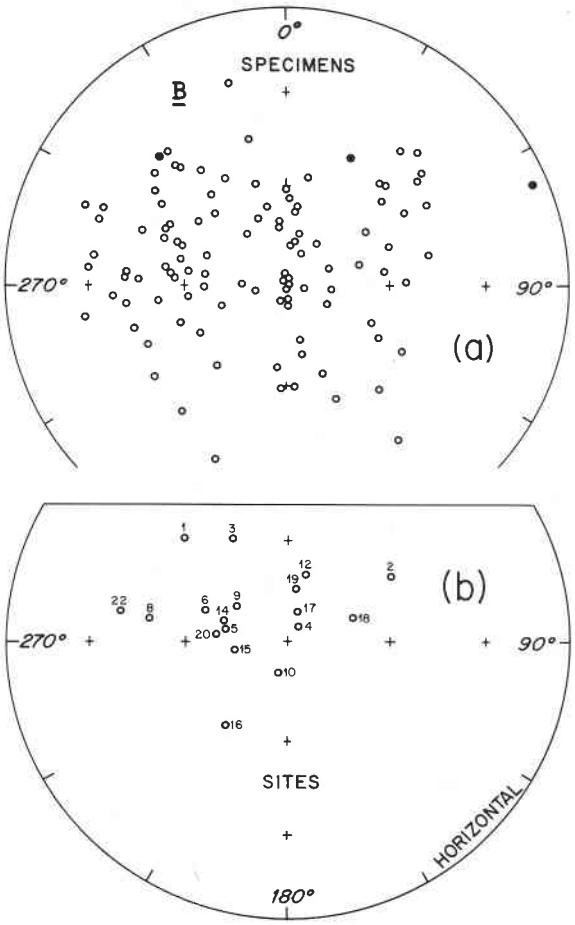


FIG. 8. Specimen (*a*) and site (*b*) *B* directions plotted. Open (solid) circles are plotted on the upper (lower) hemisphere.

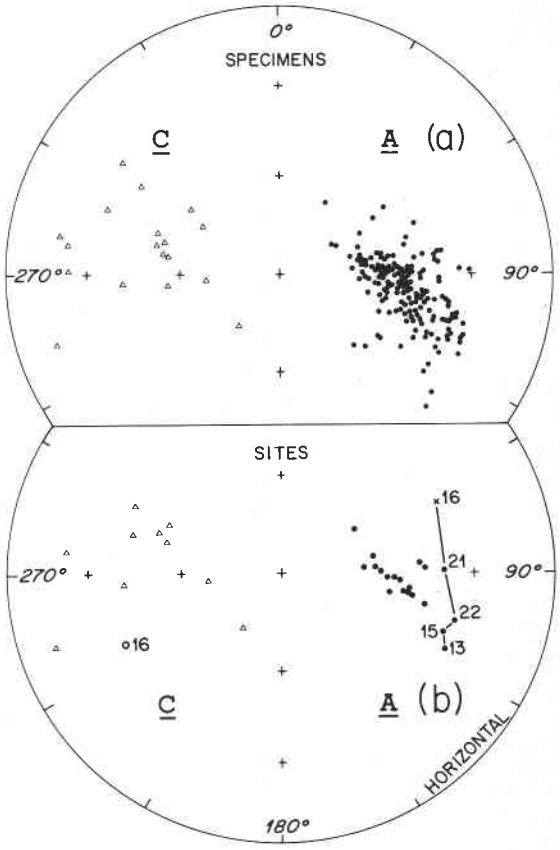


FIG. 9. *A* and *C* specimen (*a*) and site (*b*) directions plotted: *A* as circles and *C* as triangles. Open (solid) symbols are plotted on the upper (lower) hemisphere. The directions of sites 13, 15, 22, 21, and 16 (reverse) in (*b*) show a trend that reflects the positional order of these five northernmost sites from north to south on the Lake Melville shore (Fig. 2).

and *C* have similar directions, as do *B* and *D*. But what is the sequence of their acquisition? And which of them, if any, is the primary or initial magnetization?

We suggest that *A* predates *B*. This is supported by: (1) the complementary nature of the rcf and T_{UB} spectra of the *A* and *B* magnetizations in some specimens, *A* being more stable; and (2) the streaking of the *B* directions toward *A*, but not vice versa.

The evidence of (1) almost certainly indicates a partial remagnetization of an original *A* magnetization, brought about either by later heating and cooling or, more likely, by slow continuous or discontinuous cooling. These are processes that would only have affected the grains with lower T_{UB}'s or rcf's, as observed.

The streaking of the *B* directions toward *A* has at least two explanations, both of which are consistent with the presence of a partial thermoremanent magnetization (pTRM). The first explanation (*a*) is that the magnetizations were reset at different times during a period of apparent polar wander (APW). The second (*b*) is that the magnetic directions were biased owing to magnetic interactions. Explanation *a* is possible if dykes cooled through the lower range of their magnetic T_{UB} spectra at widely different times, because either they had quite different burial depths or the regional cooling regime was affected by faulting and differential uplift or by long-term local heat sources. The inland dykes are some 540 m higher in elevation than those on Lake Melville and are separated from them by a fault; therefore the relative burial depths of the two dyke sets are unknown. However, within each fault block, despite the fact that dyke sites are located within about 150 m vertically on one another, the respective directions are similarly dispersed in a non-systematic manner. A pTRM, partly chemical in nature (i.e., a pTCRM), might result in a non-systematic distribution of directions, depending on the chemical conditions at the sites. But there is no

TABLE 2. *B* site directions

Site	*N* (*n*)	*D*°, *I*°	*R*	α_{95}°
1	3 (2)	315, −47	2.726	—
2	3 (2)	057, −54	2.963	—
3	3 (2)	333, −55	2.865	—
4	4 (1)	034, −85	3.672	32
5	4 (1)	280, −72	3.662	33
6	2 (3)	291, −64	1.942	—
7	1 (4)	(132, −33)	—	—
8	4 (1)	279, −49	3.846	21
9	5	305, −72	4.718	21
10	5	197, −80	4.741	20
11	2 (3)	(340, −58)	1.312	—
12	2 (3)	015, −70	1.970	—
13	2 (3)	(057, −14)	1.845	—
14	5	287, −71	4.165	39
15	5	259, −74	3.725	52
16	5 (1)	216, −59	4.491	29
17	4	021, −81	3.267	52
18	5	070, −70	4.602	25
19	5	009, −75	4.682	22
20	5	274, −69	4.598	26
21	2 (3)	(335, −77)	1.689	—
22	2 (3)	280, −40	1.938	—

NOTES: *N* = the number of samples from which the site direction was calculated; (*n*) = the number of samples from which no direction could be determined; *D*° and *I*° = the declination and inclination of the direction; *R* = the vector resultant; and α_{95}° = the radius of the circle of confidence about the direction. Of the 78 samples with suitable data, 44 yielded data from both specimens. Those directions in parentheses were not used in calculating the grand average given in Table 5.

TABLE 3. *A* site directions and poles

		Directions			Poles
Site	*N* (*n*)	*D*°, *I*°	*R*	α_{95}°	Long.°, Lat.°
1	5	080, +62	4.968	7	171W, 39S
2	5	097, +51	4.989	4	174W, 21S
3	5	102, +46	4.996	3	175W, 15S
4	5	100, +57	4.962	8	180W, 24S
5	5	100, +50	4.979	5	176W, 19S
6	5	099, +51	4.994	3	175W, 20S
7	5	092, +58	4.983	5	176W, 29S
8	5	086, +48	4.963	7	164W, 25S
9	3 (2)	086, +62	2.986	—	176W, 36S
10	5	098, +52	4.981	5	176W, 31S
11	5	086, +65	4.988	4	179W, 38S
12	5	093, +55	4.986	5	174W, 26S
13	5	115, +35	4.980	5	180W, 02S
14	—	— —	—	—	— —
15	5	111, +37	4.985	5	178W, 05S
16	3 (3)	246, −37	2.835	—	142W, 31S
17	5	088, +46	4.960	8	165W, 23S
18	3 (2)	089, +60	2.960	—	176W, 32S
19	5	092, +56	4.993	3	174W, 28S
20	3 (2)	058, +64	2.969	—	161W, 53S
21	5	090, +40	4.989	4	163W, 19S
22	5	106, +34	4.992	4	172W, 06S

NOTE: See Table 2 for explanation of symbols.

TABLE 4. *C* site directions

Site(s)	*N*	*D*°, *I*°	*R*
1–8	—	— —	—
9	3	293, −53	2.926
10	1	285, −54	—
11	1	285, −43	—
12	1	289, −51	—
13, 14	—	— —	—
15	2	296, −42	1.942
16, 17	—	— —	—
18	1	266, −42	—
19	1	218, −70	—
20	1	265, −68	—
21	1	252, −15	—
22	3	276, −22	2.993

NOTE: See Table 2 for explanation of symbols.

evidence for quite different burial depths or for conditions affecting the regional cooling patterns. Barring a chemical effect, explanation *a* is discounted. Explanation *b*, however, is certainly viable, as is attested throughout the world by the disturbance of magnetic compass readings by magnetic anomalies associated with basic rock bodies. One can envision a process whereby unblocked low-coercive grains would be realigned in a later field that was a resultant of the external field *B* and the internal field associated with the high-coercive grains carrying the *A* magnetization (e.g., see Bailey and Hale 1982). A possible mechanism is set out and further explained in Fig. 10. Note that a field biased according to this mechanism would not necessarily confine resultant directions to one side of the *B* field direction. Also, the directions would tend to lie along a great circle passing through *A* and *B*. Thus the average directions should be largely centred on the *B* field direction. This explanation for the streaking effect, along with the other evidence, strongly suggests that *A* predates *B*. Because *C* and *D*, respectively, agree in direction with *A* and *B*, irrespective of polarity, *A* and *C* predate *B* and *D*.

Components *B* and *D* were probably acquired at about the same time, judging by their closely similar average directions. In some single specimens the two directions are distinct; this implies that in other specimens apparently containing only one low rcf or T_{UB} component

TABLE 5. Averages of site directions

		Directions				Poles	
	N	$D°, I°$	R	κ	$\alpha_{95}°$	Long.°, Lat.°	$\delta p°, \delta m°$
A	20	095.2, +52.0	19.586	46	4.9	173.4W, 22.9S	4.6, 6.7
B	18	304.9, −75.6	16.573	12	10.5	147.9E, 34.2S	17.6, 19.2
C	10	275.0, −47.7	9.335	14	13.6	170.6W, 20.1S	11.6, 17.7
C	15*	279.0, −44.8	13.951	13	10.9	171.9W, 16.1S	8.7, 13.7

NOTES: N = the number of sites (* samples); κ = the precision of the direction; and $\delta p°$ and $\delta m°$ = the semi-axes of the error ellipse about the pole, as calculated from the directions. See Table 2 for an explanation of the other symbols.

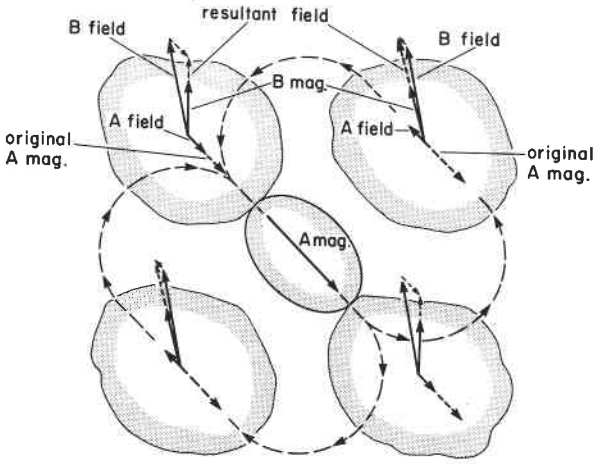


FIG. 10. Cartoon model illustrating the possible mechanism that produced the trend in the distribution of the *B* directions. The central particle represents a strongly magnetized and stable single-domain grain of titanomagnetite, which is surrounded by less strongly magnetized pseudo- and multidomain grains of the same material. These particles could also represent physically partitioned volumes within grains. Over a period of time at a raised ambient temperature the less stable grains become unblocked and develop a new direction in the prevailing external field, B. But it is suggested that this direction will be slightly biased owing to the presence of the original internal field of the stable *A* magnetization, which, since the A and B fields are not antiparallel, will tend to distribute grain directions along a great circle passing through these field directions. This "resultant" *B* magnetization will have the direction but not necessarily the intensity of the resultant field.

the direction defined as *B* may in fact be composed of *B* and *D*.

It is of some interest that there is little evidence for a component that is due to the present Earth's field (PEF) either in the rcf range above 1 mT (Fig. 6 *a* and *b*) or the T_{UB} range above 50°C (Fig. 4*b*). The *B* and *D* directions, though quite scattered, are far removed from that of the PEF. These facts show that stable components with low rcf or T_{UB} spectra can exist undisturbed in nature for hundreds of millions of years. Such apparently "soft" magnetizations might possibly be preserved by the internal fields generated by the higher coercivity components, as suggested by Halls and Palmer (1981, p. 1399). The existence of low-coercivity magnetizations with directions widely divergent from the PEF direction suggests the possibility that some stable polar directions from previous paleomagnetic studies that lie near the PEF direction, and that have been ascribed to the PEF, may in fact be legitimate ancient poles. But oppositely, very stable directions could be due to the PEF.

First magnetization

It remains to determine whether *A* or *C* is primary or initial. (Here *primary* signifies a magnetization acquired at the time of dyke crystallization and *initial* the first magnetization, whether of primary or much later origin.) One approach is to compare the metamorphic temperatures attained in the region with the highest unblocking temperatures of the *A* and *C* magnetizations, and thus determine whether the former have been high enough to completely reset the latter. There are several complications involved with this approach. First, the magnetic minerals are subject to the phenomenon of magnetic viscosity (Pullaiah *et al.* 1975) by means of which lower regional temperatures over an extended time can affect magnetizations with higher unblocking temperatures. Second, the magnetization has probably not been acquired as a simple TRM: it is likely a combined TRM and CRM—a TCRM or thermochemical remanent magnetization. This means that the effective temperature at which the magnetization was acquired is lower than the measured T_{UB}. There is no firm evidence that TCRM exists, but it can be postulated on logical grounds (Carmichael and Nicholls 1967; Grommé *et al.* 1969; Merrill 1975; Morris and Roy 1977; Roy and Lapointe 1978) and from experimental studies (Buddington and Lindsley 1964). The experimental studies have shown that magnetic mineral

systems undergo evolution and re-equilibration with decreasing temperature.

Presently available evidence suggests that subsequent to intrusion and crystallization of the Mealy dykes the terrane was either maintained, or subsequently reheated, above about 300°C but not as high as 535°C (closure temperature for radiogenic argon diffusion in biotite is 300 ± 25°C, in hornblende is 535 ± 45°C; Harrison 1981; Dallmeyer *et al.* 1981).

Would a temperature of 500°C affect the highest T_{UB}'s of magnetite by reason of the phenomenon of magnetic viscosity? Theoretical blocking curves for magnetite indicate that a temperature of 500°C held for 100 Ma would reset magnetizations with T_{UB}'s up to 545°C (Pullaiah *et al.* 1975). However, experimental evidence (Pullaiah *et al.* 1975) suggests that heating at 450°C for 10 days, corresponding to about 385°C for 10^6 years, would be sufficient to completely reset magnetite magnetizations. Pullaiah *et al.* (1975) also suggested from a compromise between theory and experiment, and taking likely chemical changes into account, that "very little, if any, remanence can survive into amphibolite facies." This implies that the magnetite remanence of the Mealy dykes has been reset.

A paleomagnetic study of the host rocks appears to verify this conclusion. Although the study is far from complete, one of the magnetic directions that is revealed by thermal treatment is identical, irrespective of polarity, to *A* and *C*. The magnetization has T_{UB}'s up to 580°C, and is likely carried by magnetite. Since the Mealy Mountains complex is several hundred million years older than the dykes (Table 1), and the magnetization is unlikely to be associated with heating due to dyke intrusion, the magnetization must represent a regional effect postdating dyke cooling. There is no evidence for a heating event at a later time to reset the magnetization, but rather for a very long, slow cooling history from the time of intrusion. The likely explanation for the *A* and *C* magnetizations and for the host magnetization is that they represent a time at which the magnetizations in both rock units became stabilized. Thus they would not represent reset magnetizations or primary magnetizations but rather the first magnetizations retained by the respective rocks. Up to this time the mineral systems would be continually evolving and re-equilibrating until the temperature and chemical conditions were conducive to the retention of a stable magnetization. It is likely under these conditions that *A* and *C* were produced as TCRM's at much lower temperatures than indicated by the T_{UB}'s.

Components *B* and *D* would have been acquired later at a lower temperature. The trend in *B* has been explained as an effect due to the magnetic influence of *A*. But whether this influence could account for all the scatter in *B* is unknown. The effect may be partly due to resetting of magnetizations at different times, depending on the chemical conditions at a site. This seems reasonable in view of the evidence that *A* and *C* are likely TCRM's produced under similar slow cooling conditions. This is supported by the overlap of the *A* and *B* T_{UB} spectra in some cases, which could point to a slight chemical effect due to grains with similar T_{UB}'s having been produced at different times and therefore in different directions.

The reason for the correspondance of the *B* and *D* directions is unknown. Pyrrhotite, which likely carries *D*, would not have acquired a magnetization at the temperature in which *B* was probably acquired (above 100°C).

Linear trend in A

The *A* magnetization, like the *B*, shows very pronounced streaking (e = 0.93; Fig. 9), which is not directed towards *B* (Fig. 11). This effect does not in general appear to be systematic as if dependent on dyke location. However, a separate trend of directions is revealed on the stereoplot (Fig. 9) within the northernmost sequence of dyke sites 13, 15, 22, 21, and 16 along Lake Melville. This trend and the main streaking direction also cannot be caused by contamination by another component. Three possible explanations for the *A* distribution are that it indicates: (1) the magnetic influence of the *C* field; (2) a lack of structural correction to the data; or (3) a period of APW. None of these possibilities is mutually exclusive.

The possibility that *C* has influenced *A* as *A* may have influenced *B* is unlikely, because *A* and *C* have nearly the same direction.

The second possibility involving structural correction is plausible because the dykes are in close proximity to a presently defined and possibly long-standing tectonic zone (Grenville Front Tectonic Zone, GFTZ), and the trend of the *A* directions is in a northwest–southeast sense, that is, perpendicular to both the dykes and the GFTZ. In addition, in Neohelikian times the region was a zone of possible incipient rifting (Baragar 1977, 1981; Meyers and Emslie 1977). The possible presence of tilting could imply rotation along block faults parallel to the dykes or a rift-related scenario (Nielsen 1975). Rifting has been invoked to explain divergent dyke attitudes in eastern Greenland (Nielsen 1975). In that region there are two co-extensive dyke swarms of different age parallel to the coast: an older swarm, which is faulted, sheared, altered, and tilted towards the margin with dips of up to 50°, and a younger, much less altered swarm, which was intruded after flexure and is interpreted to be correlative with the active spreading phase. In the present study there is no evidence for two distinct dyke phases, significant alteration, shearing, or faults to support this scenario. According to the first

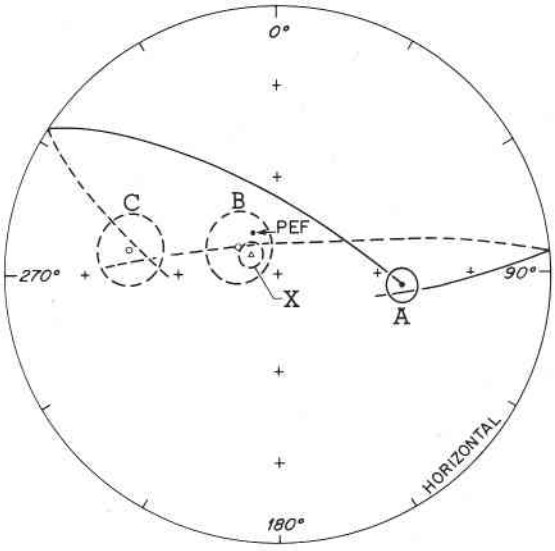


FIG. 11. The average *A*, *B*, and *C* site directions are plotted along with their respective error circles (α_{95}). The great circles representing the trends of the distributions of site directions are also shown. The *B* distribution has an average direction that lies close to the average (*X*) of the Morin complex (Irving *et al.* 1974*b*; see 482, Fig. 12), the Allard Lake anorthosite (Hargraves and Burt 1967; see 130, Fig. 12), and the Haliburton intrusions (Buchan and Dunlop 1976; see 649, Fig. 12), and a great circle that passes close to the average *A* direction. Both these facts, together with the complementary nature of the rcf and T_{UB} spectra of the *A* and *B* magnetizations, suggest that an original *A* magnetization was partially reset during the period of the Grenville regional metamorphism represented by *X* (Irving *et al.* 1974*a*). See Fig. 11 for the possible mechanism that produced the trend in the *B* distribution.

scenario, some or all of the dykes may have been tilted about a horizontal axis parallel to the dykes by local doming or downwarping following or accompanying intrusion. Later dykes might then be injected into younger fractures somewhat rotated relative to older fractures. The idea of tilting is currently being tested by measuring the magnetic anisotropy, the rationale being that emplacement directions, which can be determined by anisotropy of magnetic susceptibility, should define dyke altitudes (Ellwood 1978).

The third possibility of APW is consistent with the fact that the individual site poles are aligned along the Keweenawan track (Figs. 12, 13). We discounted APW as an explanation for the streaked *B* directions if *B* is a pTRM, because there is no evidence that dykes cooled at widely different times. The same would apply for the *A* directions. However, if *A* is partly chemical in nature (a TCRM), it may have been acquired at different times because of differing chemical conditions at each site.

Poles

The Mealy dyke *A* and *B* poles, Md(A) and Md(B), are plotted in Fig. 12. Md(B) is secondary and lies close to the Grenville A poles marking the Grenville regional event (poles 130, 482, 649). These poles have been dated at 980 Ma by $^{40}Ar/^{39}Ar$ (Berger *et al.* 1979).

Md(A) represents an earlier magnetization. It does not lie on the defined Grenville track, but rather on a simple connection between the Grenville and Keweenawan tracks. Its position suggests that it may either relate to the Keweenawan Track or to an undefined path that perhaps joins the early part of the Logan Loop at the equator (Fig. 13). Md(A) certainly does not relate to the 1300–1400 Ma poles that relate to the time of Mealy dyke intrusion (1380 ± 54 Ma) (Fig. 12). Our interpretation is that *A* was acquired during late Keweenawan time, much later than the time of dyke intrusion. This is in accord with the evidence presented earlier of an extensive period of re-equilibration of the magnetic minerals.

Paleomagnetic poles derived from rock units that are probably related to the Mealy dykes are shown in Fig. 12 (Sg, Mg, Sl, Hd: shaded). Only Sg shows some agreement with Md(A). It has been suggested that Sg is primary (Fahrig *et al.* 1974), but its large error indicates that the associated magnetization could be a resultant of several components. Irving *et al.* (1977) observed that the three presumably unmetamorphosed Sg sites located north of the Grenville Front have an average pole some 30° east of the pole representing the remaining Sg sites, in close agreement with 1300–1400 Ma poles (Fig. 12), thus supporting the possible mixed nature of the Sg magnetization. Of the poles from other related units, Mg (Fahrig and Larochelle 1972) is considered to be secondary (Fahrig and Loveridge 1981), probably related to the later Hadrynian polar track represented by the Grenville G(B) poles (Fig. 13). Sl and Hd, on the other hand, may represent primary magnetizations (Park and Roy 1979; Irving *et al.* 1977). They have respective apparent ages of 1323 ± 92 and 1316 ± 94 Ma (Table 1). The same general directions and relations are represented by the upper and lower sequences of the similarly dated (1282 Ma) Gardar lavas of southern Greenland (Piper 1977), which are possibly related to the Seal and Harp rocks (Baragar 1981).

Poles 415 and 417, derived from an earlier paleomagnetic study of the host rocks of the Mealy dykes (the Mealy Mountains complex), lie at opposite ends of the linear distribution of Md(A) site poles (Fig. 12) (Fahrig *et al.* 1974). Each has a rather large error. In the earlier work, samples were only treated with AF's. It is therefore possible that the errors reflect several unresolved magnetizations, perhaps consisting of normal and reverse components. These results are in contrast to those indicated from a current partial study of the Mealy

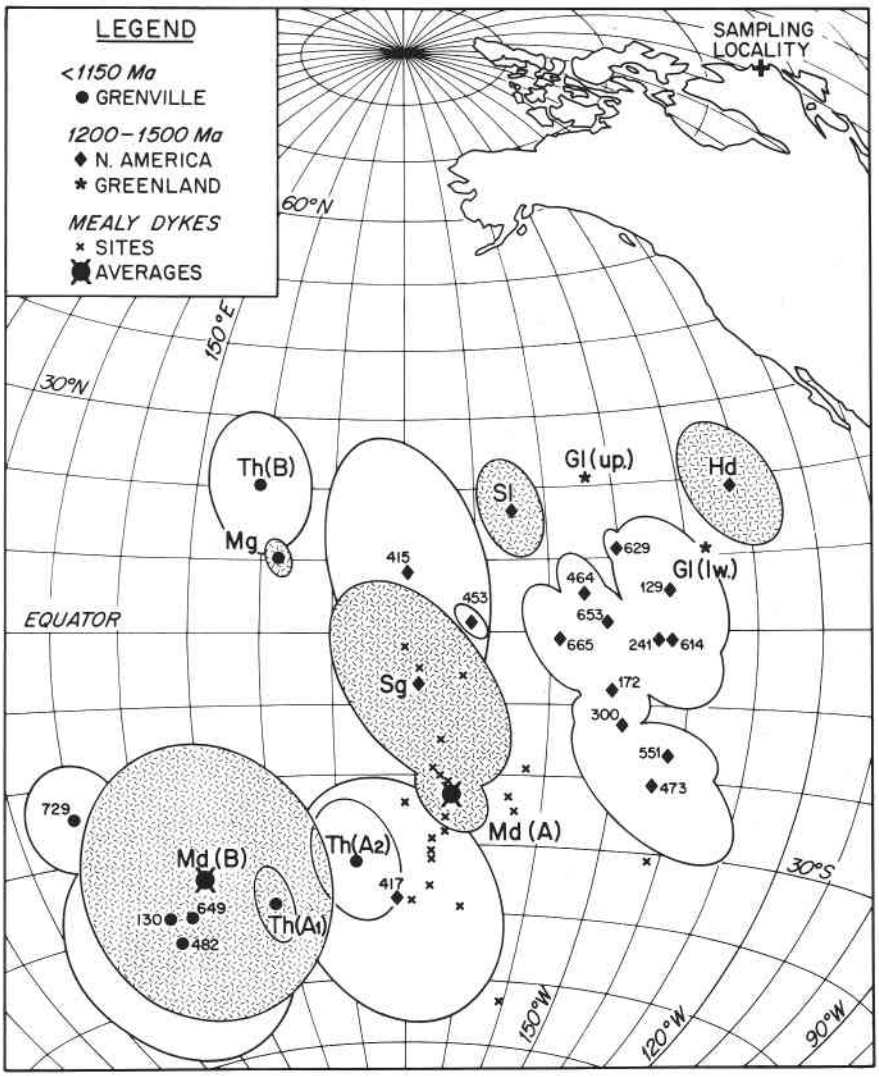


FIG. 12. Mealy dyke poles Md(A) and Md(B) (large symbols, shaded ellipses) plotted with reference to other poles from supposedly related rocks of the Grenville and Nain structural provinces (small symbols, and shaded ellipses) and to pertinent pre-Grenville poles of the North American craton and Greenland. Ellipses denote the polar errors (δp, δm). Numbers refer to poles found in the following listings: Irving (1964), Irving and Hastie (1975), and an updated version of the latter (not published). The poles grouped according to the legend on the figure are as follows. *Grenville–Nain*: Mg = Michael gabbro (Fahrig and Larochelle 1972); 130 = Allard Lake anorthosite (Hargraves and Burt 1967); 482 = Morin complex (Irving *et al.* 1974*b*); 649 = Haliburton intrusions, A (Buchan and Dunlop 1976); 729 = Magnetawan metasediments (McWilliams and Dunlop 1975); *1200–1500 Ma (N. America)*: Hd = Harp dykes (Irving *et al.* 1977); Sg = Shabogamo gabbro (Fahrig *et al.* 1974); Sl = Seal Lake Group basalts and gabbros (Park and Roy 1979); 129 = Croker Island complex, Ontario (Palmer 1969); 172 = Sherman granite (Eggler and Larson 1968); 241 = St. Francois rocks (Chi-hsu *et al.* 1966; Hays and Scharon 1966); 300 = Belt Series combined (Collinson and Runcorn 1960; Norris and Black 1961; Black 1963; Runcorn 1964); 415 = Mealy Mountains anorthosite, northwest component (Fahrig *et al.* 1974); 417 = Mealy Mountains anorthosite, east component (Fahrig *et al.* 1974); 453 = Mackenzie dykes combined, Interior Laurentia; 464 = Seal Lake Group, red beds (Roy and Fahrig 1973); 473, Sibley Group red beds, Ontario (Robertson 1973); 551 = Belt Series, Alberta–Montana (Evans *et al.* 1975); 614 = Michikamau intrusion (Murthy *et al.* 1968; Emslie *et al.* 1976); 629 = Nain anorthosite complex (Murthy 1978); 653 = Harp Lake complex (Irving *et al.* 1977); 665 = Mistastin pluton (Fahrig and Jones 1976); *Greenland*: Gl (lw.) = Gardar lavas, lower (Piper 1977); Gl (up.) = Gardar lavas, upper (Piper 1977). The Gardar poles have been corrected for the subsequent relative rotation of Greenland with respect to North America (Piper 1977).

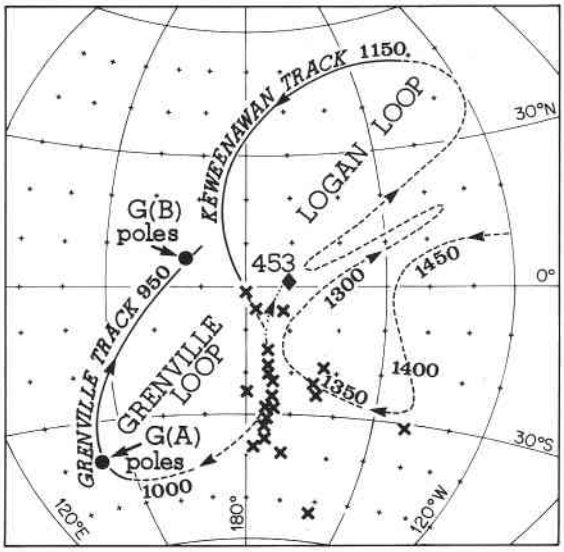


FIG. 13. The Mealy dyke site poles (crosses) compared with a schematic representation of a possible APW path of Laurentia for the approximate period 1500–900 Ma. The depicted path is in part after Roy and Robertson (1978), Irving and McGlynn (1981), and Halls and Pesonen (1982). The dashed segments of the curve are tentative. The dotted curve is an alternate possibility suggested by the results of this paper. Pole 453 is explained in Fig. 12.

Mountains complex obtained by thermal demagnetization. In this study both normal (3 specimens; $D = 096°$, $I = +52°$; $\kappa = 67$) and reverse (10 specimens; $D = 272°$, $I = -53°$; $\kappa = 14$) directions from 13 sites agree to within 3°, and their combined pole (172°W, 25°S) virtually agrees with Md(A) (173°W, 23°S). The magnetization has T_{UB}'s between about 350 and 580°C, indicating that it probably resides in magnetite.

The Md(A) and Md(C) poles, together with the poles from the host rocks (415, 417), probably represent the oldest magnetizations so far obtained from the Grenville Province. They "see through" the 1000 Ma Grenville event. The only other study that may indicate an older pole was conducted on the Thanet gabbro of the Hastings Lowlands of southeastern Ontario (Buchan 1978). The three poles ($Th(A_1)$, $Th(A_2)$, Th(B)) derived from that study are plotted in Fig. 12. Only $Th(A_1)$ appears to be a TRM. It has been dated at between 1000 and 1050 Ma by $^{40}Ar/^{39}Ar$ (Berger and York 1981). The other two poles are considered to be CRM's, probably acquired at low temperatures somewhat later. However, Berger and York noted that, "On the available evidence the possibility that Th(B) was acquired near 1.2 Ga cannot be excluded." But they also indicated that Th(B) plots on a possible younger extension of the Grenville track at about 800 Ma.

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